OLIST E-COMMERCE
BUSINESS ANALYTICS

› Purinut Chairungrueang

Technical Methodology

01

Preprocessing:
Cleaning data and constructing a Relational Database

02

SQL Analysis:
Relational data joining and initial insights.

03

Visual Storytelling:
Exploratory Data Analysis (EDA) to find preliminary patterns.

04

Statistical Modeling:
RFM Analysis and Hypothesis Testing (T-tests).

05

BI Deployment:
Final Executive Dashboard (Power BI).

[01] Data Preprocessing & Database Ingestion

- **Temporal Calibration**: Converted raw `object` timestamps into `datetime` format to enable precise time-series and lead-time analysis.
- **Semantic Standardization**: Translated `product_category_name` from Portuguese to English and handled missing values by imputing them as "others".
- **Spatial Deduplication**: Cleaned the `geolocation` dataset by removing duplicate coordinates to ensure mapping accuracy.
- Database Ingestion: Constructed a central `olist.db` (Relational Database) using Python’s `.to_sql()` method to serve as the "Single Source of Truth."

[02] SQL Analysis & Strategic Insights

Key Analytical Focus:

- **Revenue Dynamics:** Monthly trends & Peak season (Nov 2017) performance.
- **Market Demand:** Top 10 vs. Bottom 10 Product Categories.
- **Logistics Benchmarking:** Identifying Geographic Delivery Bottlenecks (Slowest States).
- **Financial Landscape:** Analyzing Payment Method Distribution & Preferences.

Outcome: Validated Single Source of Truth for Exploratory Data Analysis (EDA).

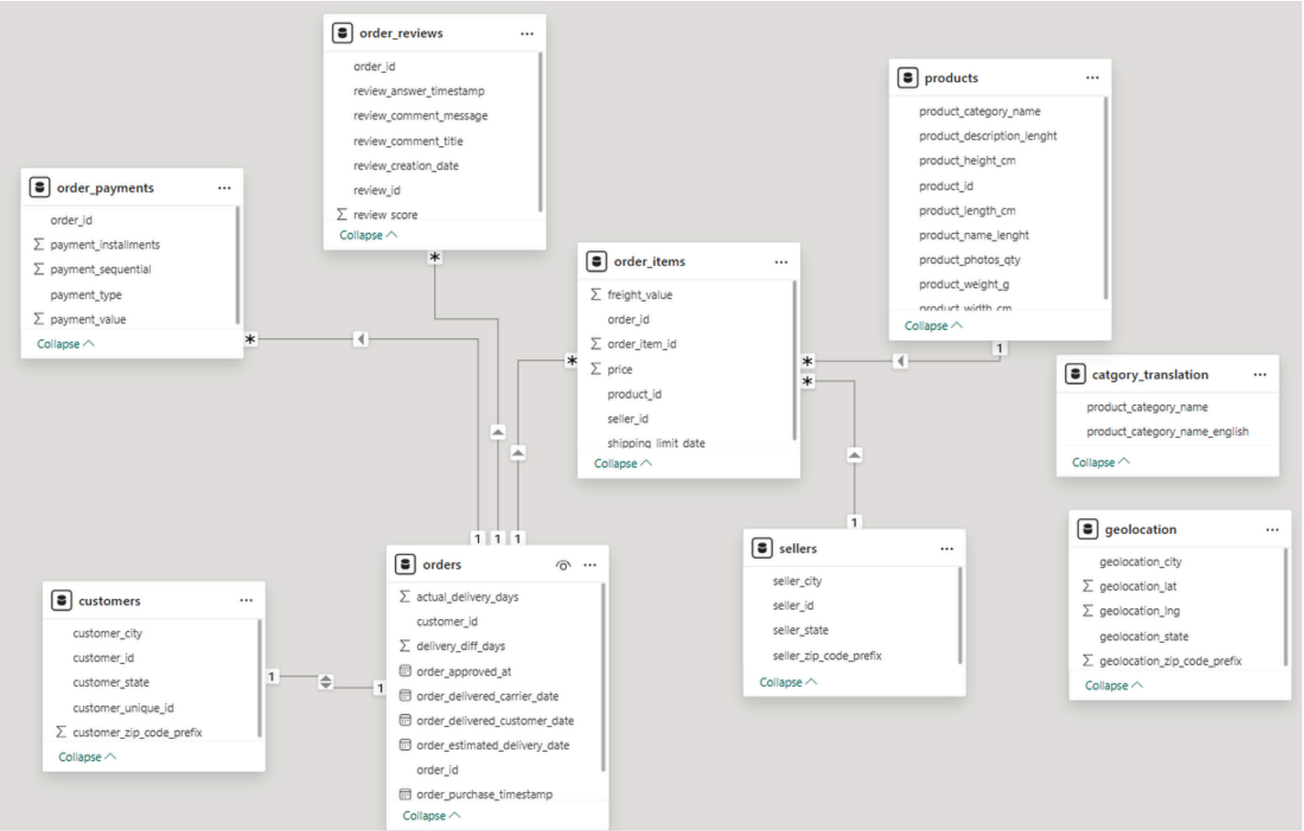
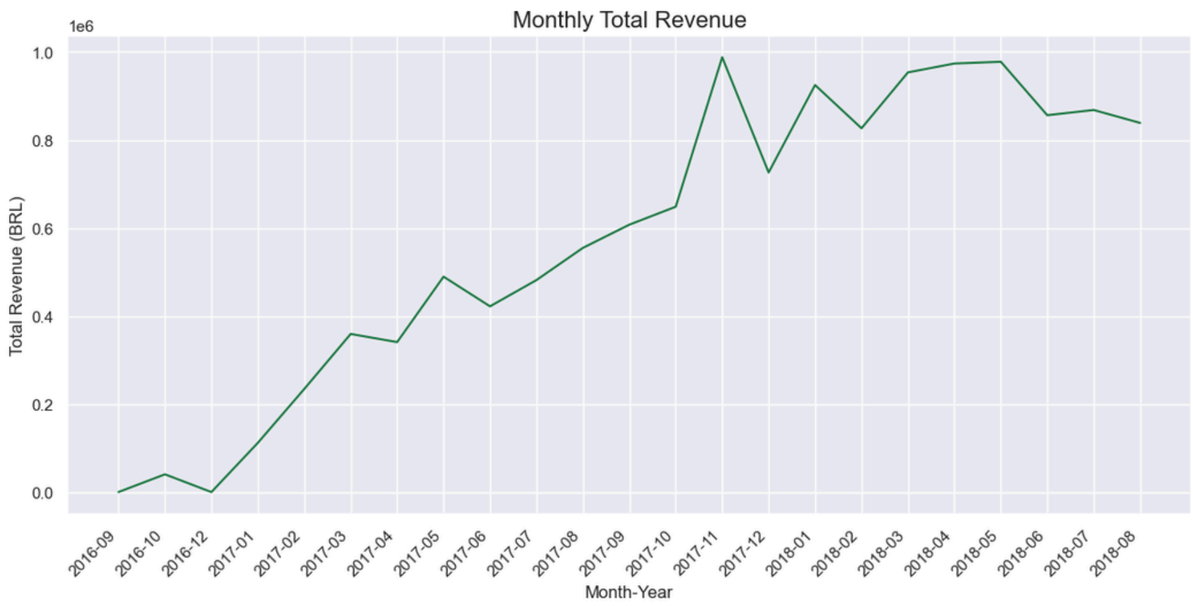


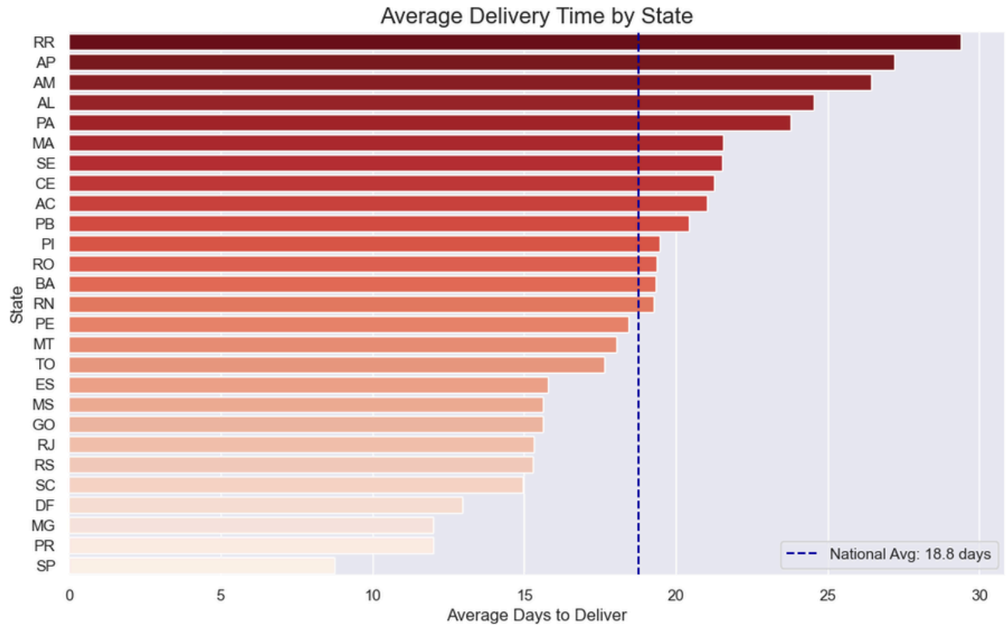
Figure: *Database Schema of Olist E-commerce Dataset*

[03] Visual Storytelling: Exploratory Data Analysis (EDA)

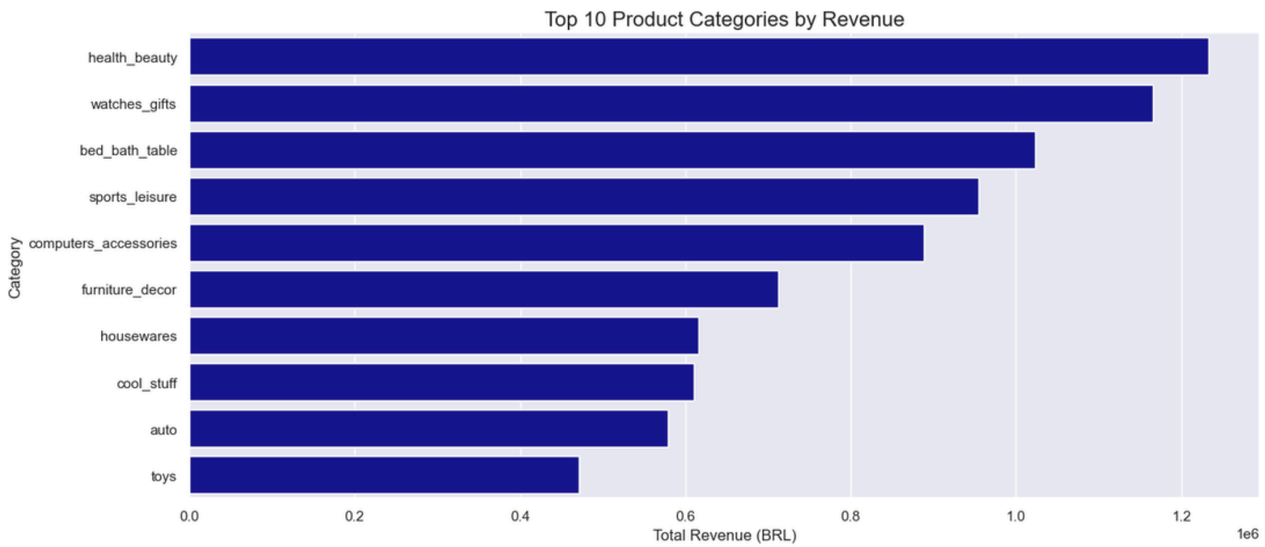
1. **Growth & Seasonality:** Analyzed Monthly Revenue Trends to monitor business scaling and detect peak performance periods.

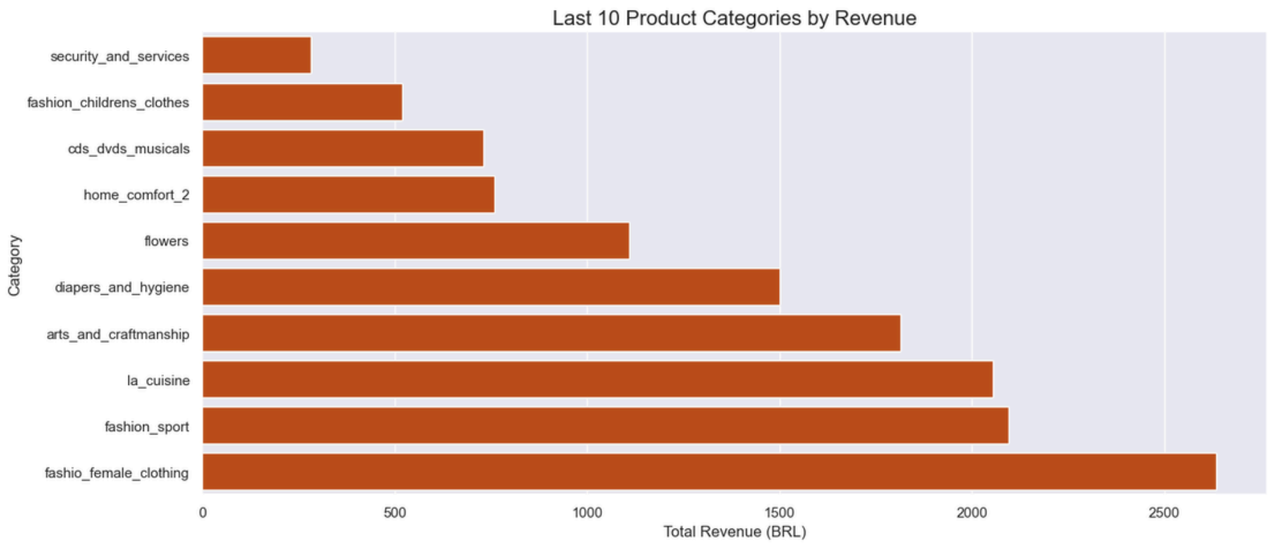


2. **Logistics Efficiency:** Visualized State-level Delivery Lead Times against the National Average, revealing significant geographic performance gaps.

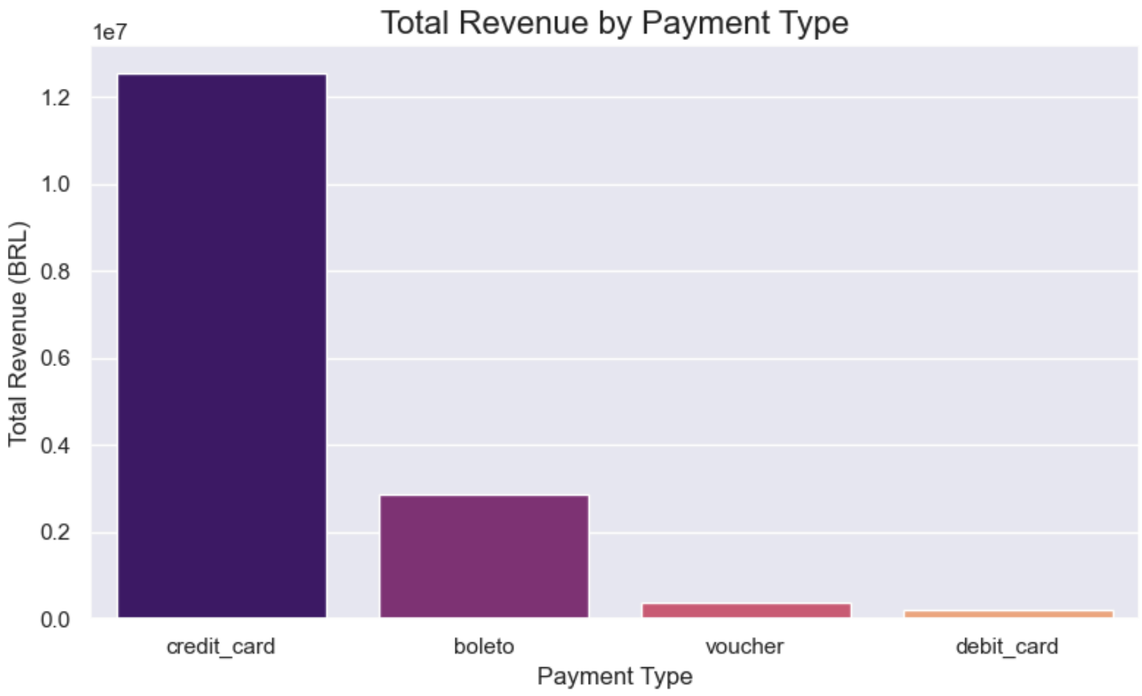


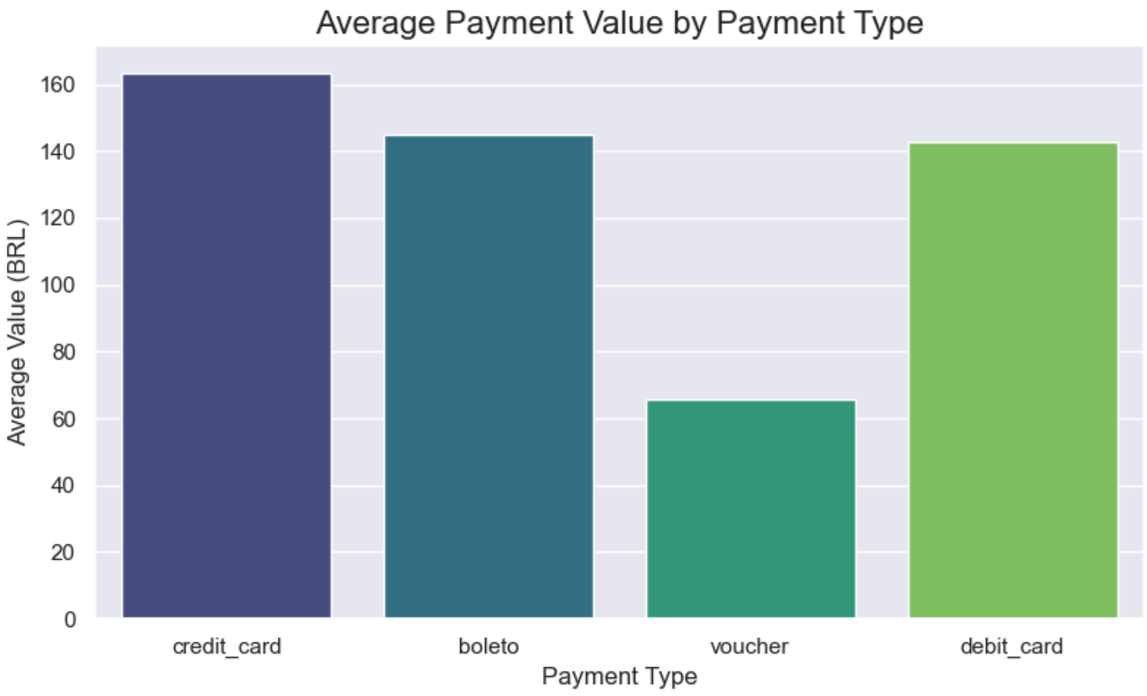
3. **Product Portfolio Health:** Benchmarked the Top 10 vs. Last 10 Categories to identify high-demand vs. inventory remaining in stock.





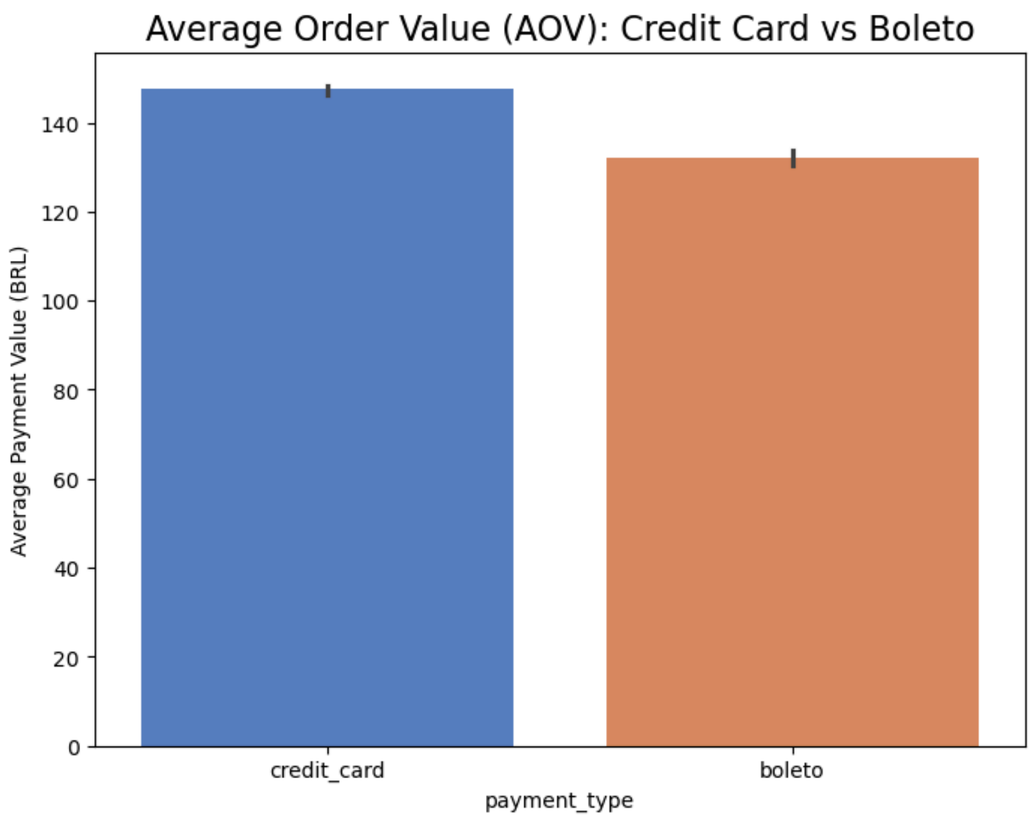
4. **Financial Dynamics**: Compared Total vs. Average Revenue by Payment Type, uncovering the disparity between transaction volume and value.





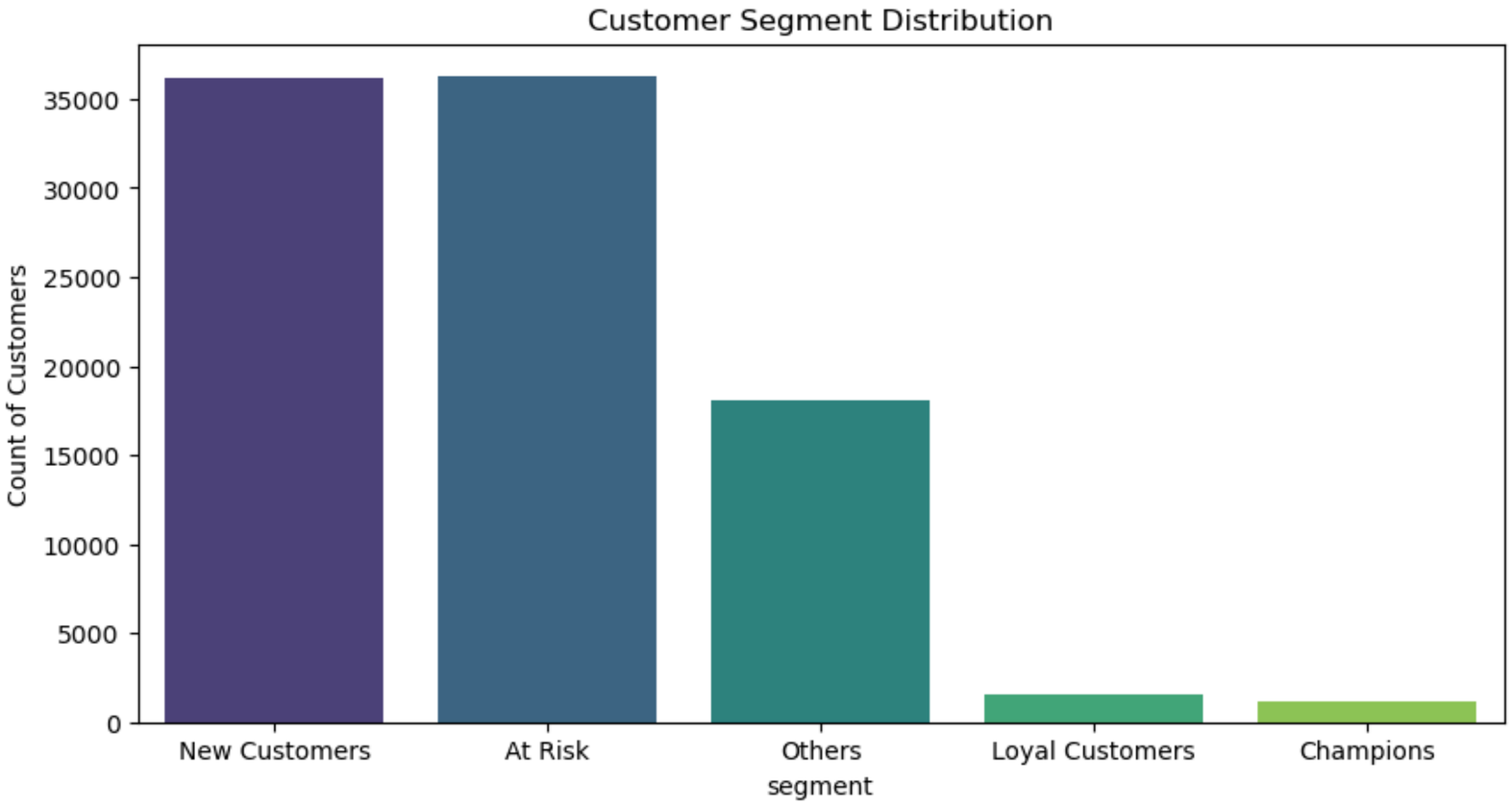
[04] Customer Behavior and Statistical Testing

1. Payment Impact (Hypothesis Testing):
   - **Method:** Independent T-test on AOV (Credit Card vs. Boleto).
   - **Technical:** Applied 99th Percentile Clipping for outlier removal, preserving high-value behavior while ensuring statistical robustness.
   - **Result:** Rejected H0 ($p < 0.001$). Credit cards significantly drive higher transaction values via installments.



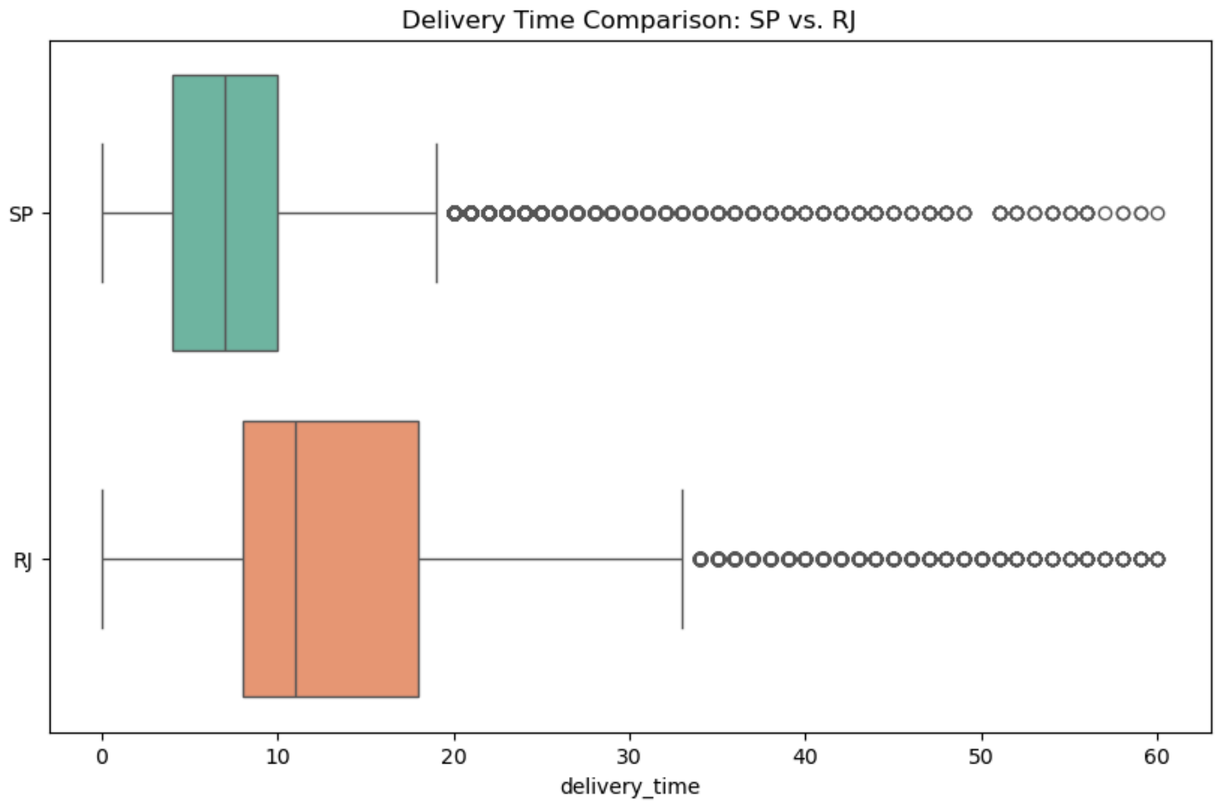
2. Behavioral Segmentation (RFM Analysis):

- **Framework:** Quintile-based scoring (1-5) for Recency, Frequency, and Monetary metrics.
- **Discovery:** Identified a high concentration of "At Risk" and "New Customers," signaling a critical need for retention-focused strategies.



3. Regional Benchmarking (A/B Test):

- **Comparison:** São Paulo (SP) vs. Rio de Janeiro (RJ).
- **Statistical Tool:** Welch's T-test (Accounting for unequal variance).
- **Insight:** Significant performance gap ($p < 0.001$). SP's logistics efficiency directly correlates with superior customer review scores.



[05] Executive Dashboard:
Strategic Power BI Deployment

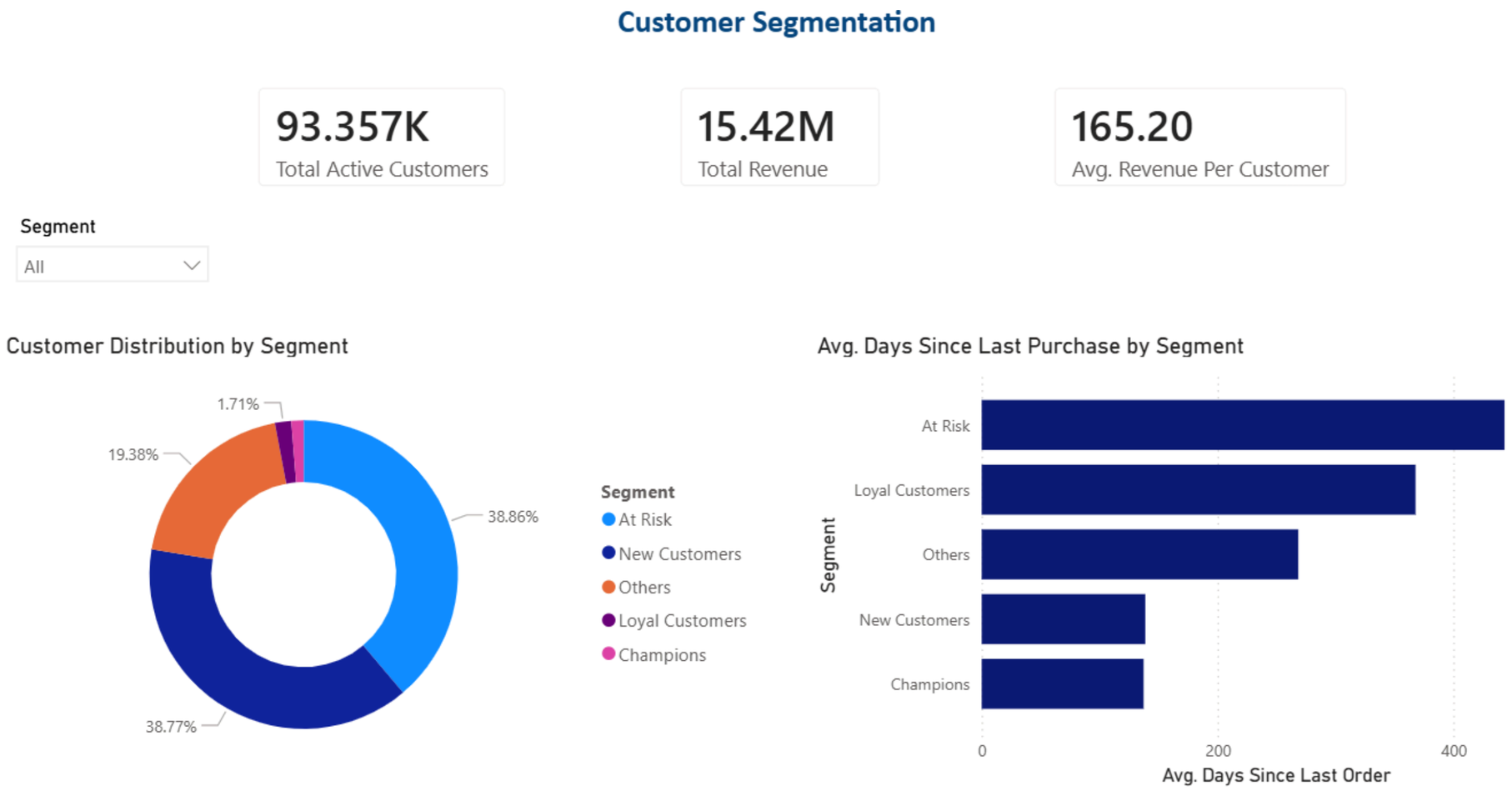
Objective: Transforming Complex Data Models into a High-Impact Decision Support System.

Tool: Developed using Power BI for cross-platform accessibility and real-time exploration.

Key Features:
Interactive Analytics: Dynamic slicers and drill-throughs for granular insights by State, Segment, and Category.

Dashboard Architecture

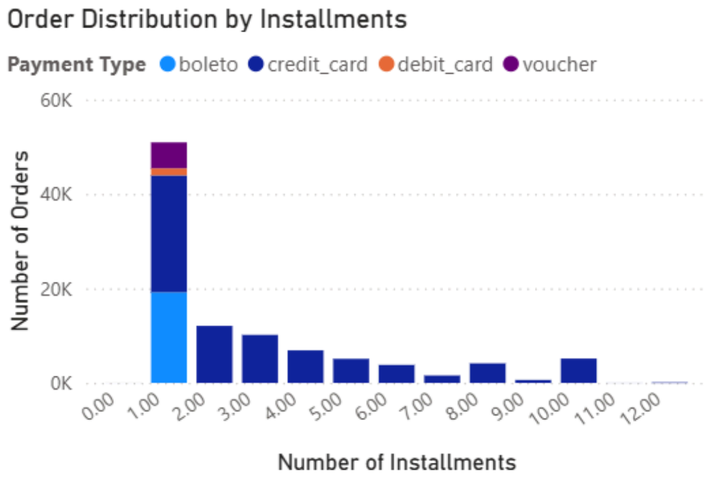
Page 1 Customer Segmentation:
Visualizing the health of the customer base and identifying high-value "Champions" vs. "At Risk" segments.



Page 2 Payment Performance:
Monitoring AOV trends and the revenue impact of installment-based credit transactions.

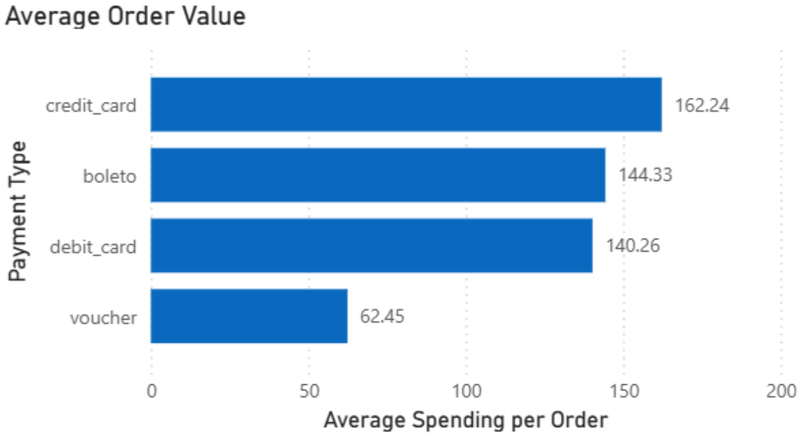
Payment Performance

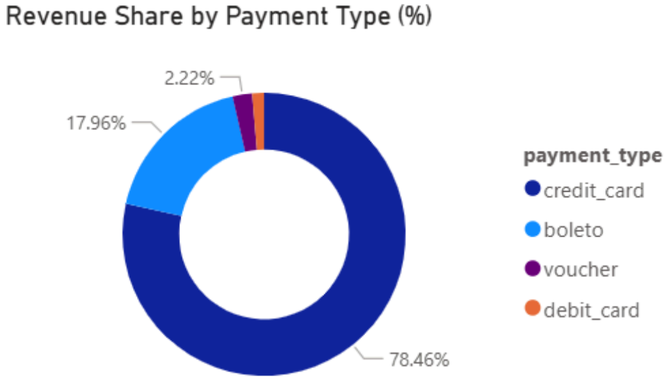
Payment Type
- boleto
- credit_card
- debit_card
- voucher



100.756K
Total Transactions

153.07
Average Order Value





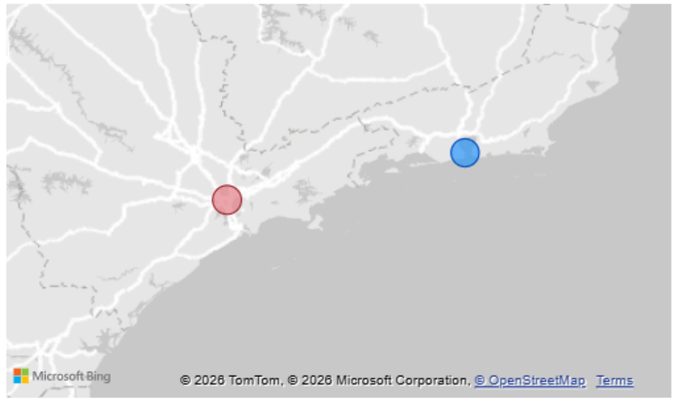
Page 3 Logistics & Regional Insights:
Spatial analysis of the SP vs. RJ performance gap, correlating delivery lead times with customer satisfaction.

Delivery Efficiency: São Paulo vs. Rio de Janeiro

9.61
Avg. Delivery Time (Days)

4.19
Avg. Customer Satisfaction

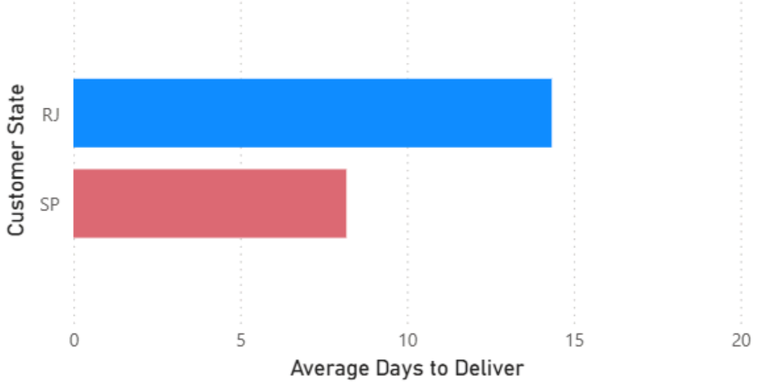
Geographic Satisfaction Map



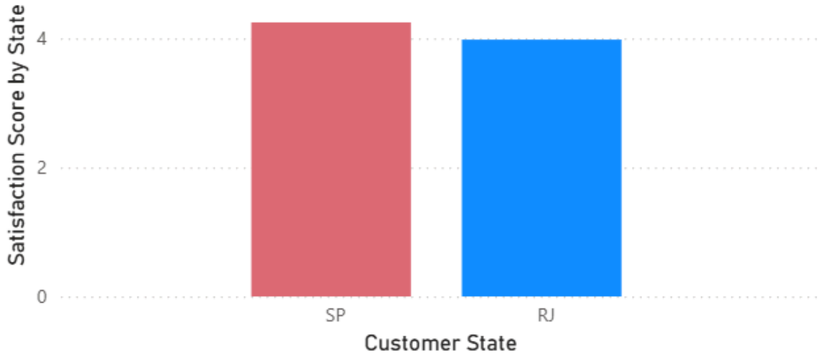
Customer State

RJ | SP

Delivery Performance by State



Satisfaction Score by State



Conclusion & Strategic Recommendations

Core Recommendations

1. **Retaining our Best Customers (Top 1.71%):** Launch a VIP Loyalty Program for "Champions" and "Loyal" customers. We need to focus on keeping this small but powerful group shopping with us for the long term.
2. **Turning New Customers into Regulars:** Use personalized offers to encourage New Customers to buy again. Our goal is to grow that small 1.71% loyal base by building better relationships with first-time buyers.
3. **Driving Higher Sales with Flexible Payments:** Promote Credit Card installments to encourage larger total purchases. Our data confirms ($p < 0.001$) that offering monthly payments is the most effective way to increase Average Order Value (AOV) across the platform.
4. **Fixing Delivery in Rio de Janeiro (RJ):** Improve shipping speed and logistics in the RJ region. Faster delivery is the direct key to stopping customers from leaving and getting better review scores.

THANK YOU

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